

3. Sleep Hours Distribution (Histogram)

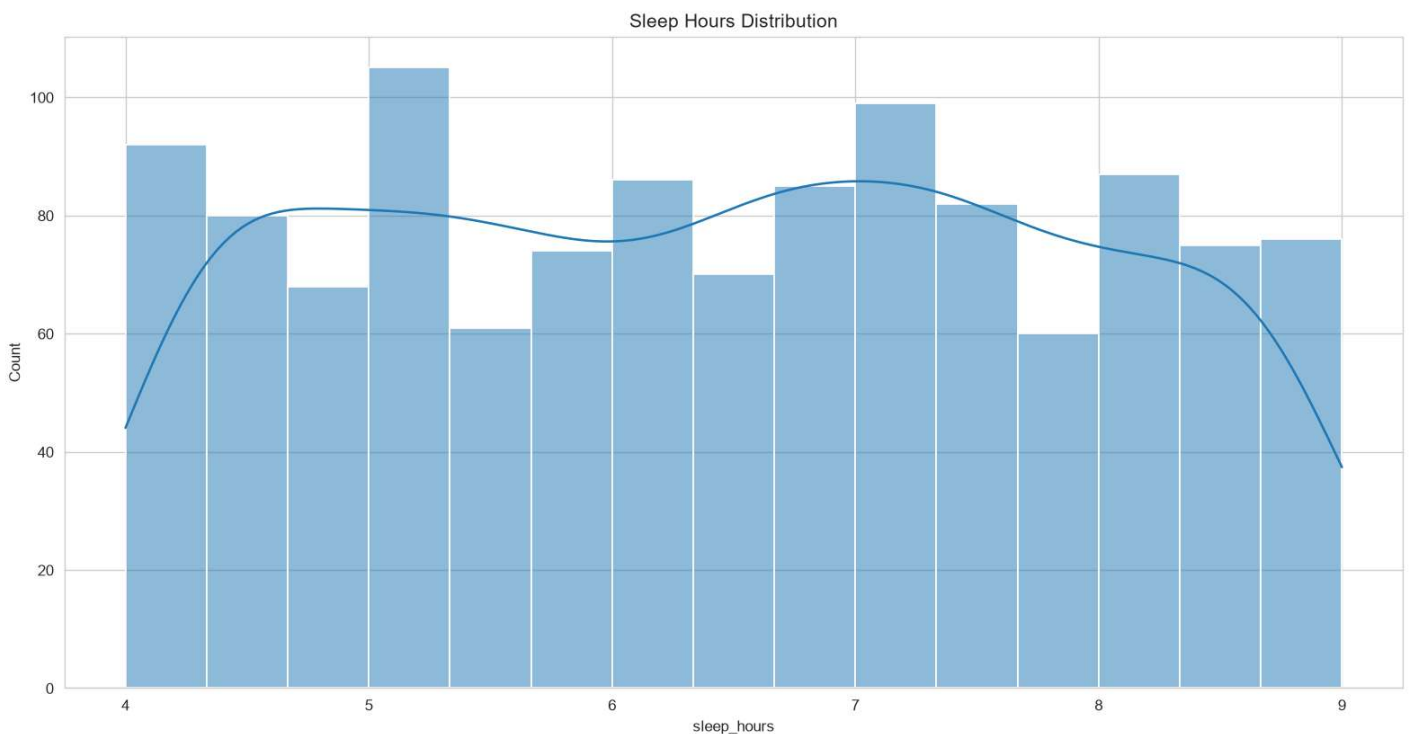
Description

The Sleep Hours Distribution histogram illustrates how daily sleep duration varies among teenagers included in the dataset. Adequate sleep plays a critical role in maintaining physical health, cognitive performance, emotional regulation, and overall mental well-being. Insufficient sleep has been associated with increased stress, anxiety, depression, reduced concentration, and lower academic performance.

Python Code

```
sns.histplot(df['sleep_hours'], bins=15, kde=True)
plt.title('Sleep Hours Distribution')
plt.show()
```

Graph



Interpretation

The histogram shows that most students sleep within a moderate range of hours, while fewer students appear at the extreme ends of the distribution. The KDE curve suggests the general pattern of sleeping behaviour among participants without relying solely on histogram bars. Understanding sleep distribution is important because sleep quality directly influences emotional stability, learning ability, and overall mental health. Although this visualization does not establish causal relationships, it provides valuable descriptive insights into teenage sleeping behaviour and creates opportunities for future analysis involving stress levels, anxiety, and academic performance.